CA Risk Monitor

Sachu Mon P. Sajeev · TSI 2026

Dashboard Mode

- Real Data — VitalDB
- Synthetic — Deep Dive

VitalDB Patients

View Mode

- Single Patient
- Compare 2 Patients

Filter by

All Patients

Try fetch live waveform (VitalDB API)

Alert threshold (%)

55

Lookback window (min)

60

TAN · AUROC 0.9937 LightGBM · AUROC 0.9073 TSI University · 2026

INTRAOPERATIVE CARDIAC ARREST RISK MONITOR

REAL DATA — VitalDB

VitalDB · 6376 Patients · CA=66 · No-CA=6310 | 12 Apr 2026 · 14:14 | Single Patient

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TAN · AUROC	LightGBM · AUROC	TAN · AUPRC
0.9937	0.9073	0.6027
95% CI [0.9909–0.9961]	60-min window	240-min window

Total: 6376 | CA: 66 | No CA: 6310 | 240min: 1731 | 120min: 4197 | 60min: 6047 | 30min: 6376

Select Patient (6376 available)

#1 · NoCA · 120min

Case #1 ✓ NO CA 120-min window Feature-Reconstructed

⚠ Only 120-min data available for this patient. Windows [240] not available. Comparison limited to 120-min analysis.

✓ 30min ✓ 60min ✓ 120min ✗ 240min

CA RISK · 120MIN WINDOW

17.1%

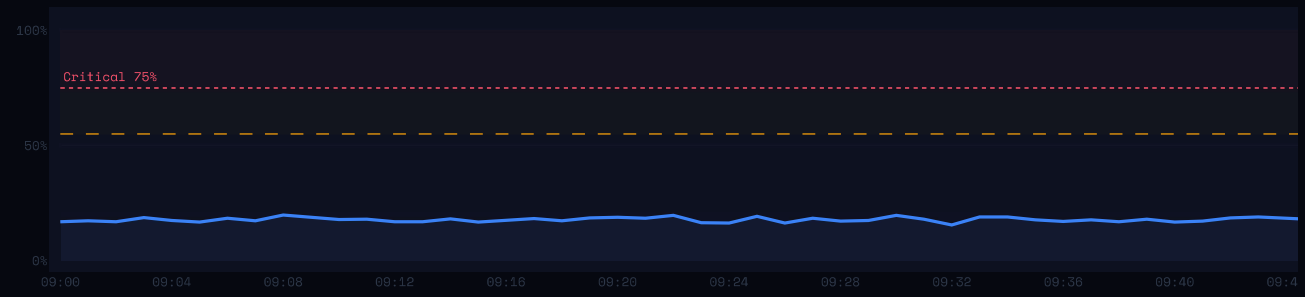
LOW

HEART RATE	SPO₂	ETCO₂	MEAN BP	SYSTOLIC BP	DIASTOLIC BP
77	100	32	94	138	63
bpm	%	mmHg	mmHg	mmHg	mmHg
→ OK	↑ HIGH	↓ LOW	→ OK	→ OK	→ OK
→0.3/30m	→0.1/30m	→0.1/30m	→0.2/30m	→0.3/30m	→0.2/30m
Range: 77–77	Range: 100–100	Range: 32–32	Range: 94–94	Range: 138–138	Range: 63–63
Normal: 50–100	Normal: 95–100	Normal: 35–45	Normal: 70–105	Normal: 90–140	Normal: 60–90

CLINICAL FEATURE STATISTICS (FROM CSV)

SIGNAL	MEAN	STD	MIN	MAX	SLOPE	NORMAL RANGE
Heart Rate	77.0	0.00	77.0	77.0	0.0000	50–100 bpm
SpO_2	100.0	0.00	100.0	100.0	0.0000	95–100 %
$EtCO_2$	32.0	0.00	32.0	32.0	0.0000	35–45 mmHg
Mean BP	94.0	0.00	94.0	94.0	0.0000	70–105 mmHg
Systolic BP	138.0	0.00	138.0	138.0	0.0000	90–140 mmHg
Diastolic BP	63.0	0.00	63.0	63.0	0.0000	60–90 mmHg

RISK SCORE OVER TIME



VITAL SIGNAL TRENDS

📊 Waveform reconstructed from CSV statistical features (mean, std, slope). VitalDB API fetch attempted automatically.

Heart Rate (bpm) · normal 50–100

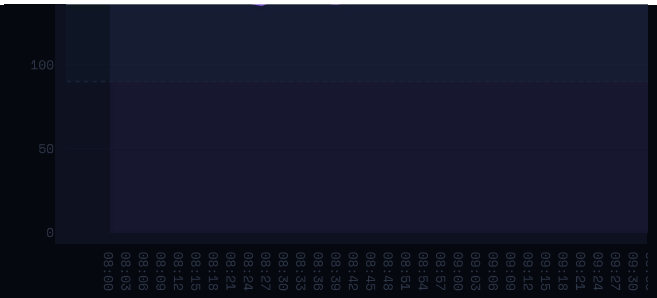
SpO₂ (%) · normal 95–100

EtCO₂ (mmHg) · normal 35–45

Mean BP (mmHg) · normal 70–105

Systolic BP (mmHg) · normal 90–140

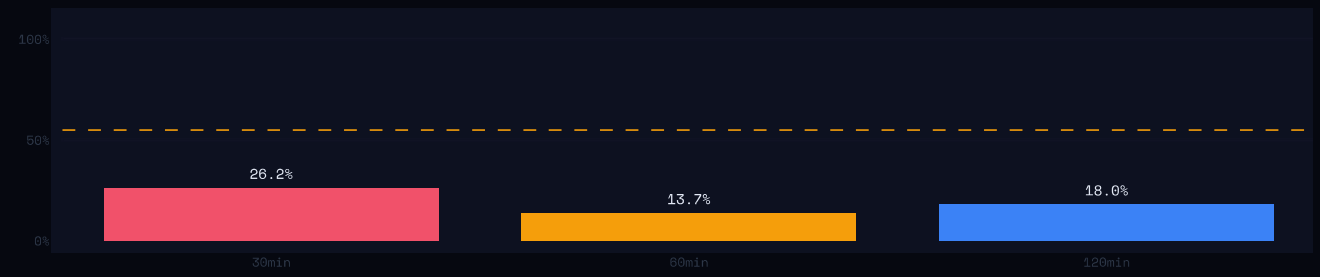
Diastolic BP (mmHg) · normal 60–90



CHANGE POINT LOG

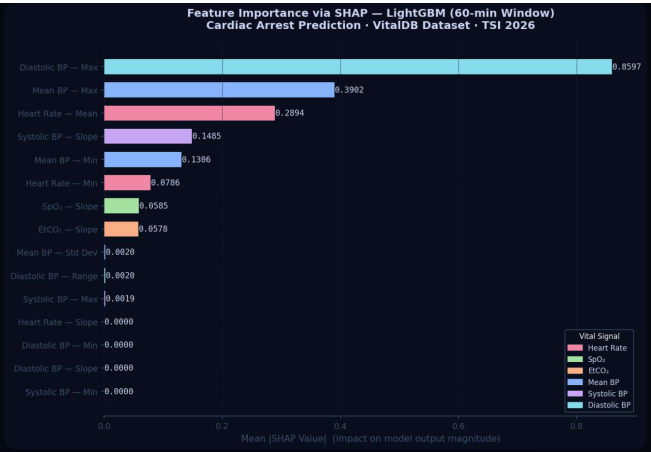
No significant change points detected.

RISK ACROSS AVAILABLE WINDOWS

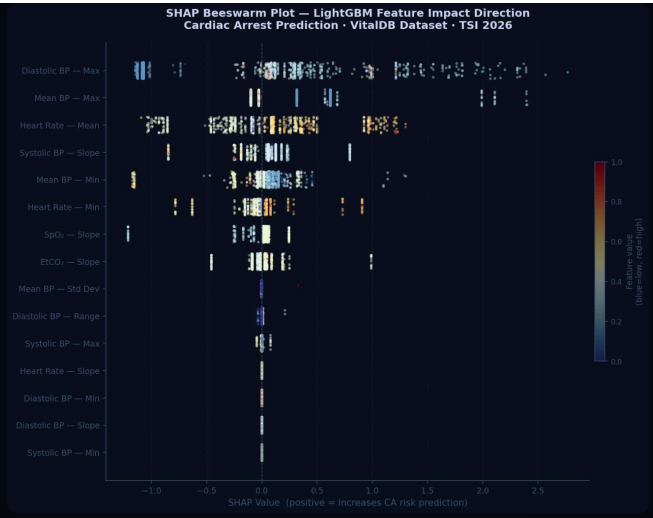


🔬 SHAP Feature Interpretability — LightGBM (60-min Window)

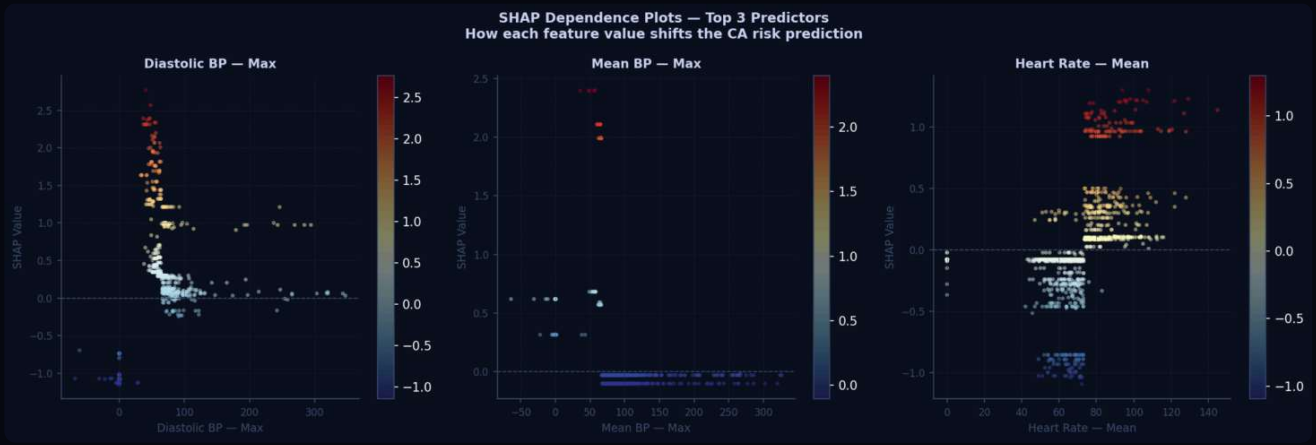
SHapley Additive exPlanations · TreeExplainer · AUROC 0.9073 · VitalDB Dataset · TSI 2026



Mean |SHAP| — Top 15 Features



SHAP Beeswarm — Feature Impact Direction



SHAP Dependence — Top 3 predictors

🔬 **How to read these charts:**

Bar chart — average magnitude of each feature's impact. Longer bar = stronger predictor.

Beeswarm — each dot is one patient. Red dots pushed right = high value increases CA risk. Blue pushed left = decreases risk.

Key finding — Diastolic BP Max, Mean BP Max, and Heart Rate Mean are the top 3 predictors. Run `python shap_analysis.py` to regenerate.

🟢 **240min** — Full 4-hour window | 🔵 **120min** — 2-hour window | 🟡 **60min** — 1-hour window | 🔴 **30min** — Minimal 30-min window

🔴 CRITICAL ≥75% — Alert physician immediately | 🟠 HIGH 50–75% — Increase monitoring | 🟡 MODERATE 25–50% — Watch closely | 🔵 LOW <25% — Routine monitoring

Clinical decision-support prototype · All treatment decisions remain with the attending physician · Sachu Mon P. Sajeev · MSc Data Science & AI · TSI University Riga · 2026